GT For Polyhedra

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1 Goldblatt-Thomason I

Definition 1. *Let P, X be polyhedra, Σ a triangulation of X, and U an open subpolyhedron of P.*
When Σ is equipped with the poset topology, we call $f : U \to \Sigma$ a ***face up-reduction*** *$P \looparrowright X$ if*

- *f is polyhedral, i.e. $f^{\leftarrow}(V)$ is an open subpolyhedron of U for any upward closed set $V \subseteq \Sigma$*
- *f is open*

Theorem 2. *A class of polyhedra $\mathcal{C}$ is modally definable if and only if it's closed under finite disjoint unions and face up-reductions, i.e.*

(I) if $P, Q \in \mathcal{C}$, then $P \sqcup Q \in \mathcal{C}$

(II) if $P \in \mathcal{C}$ and $P \looparrowright X$, then $X \in \mathcal{C}$.

Proof ($\Longrightarrow$). Let $\mathcal{C}$ be a modally definable class, i.e. $\mathcal{C} = \left\{ P \in \mathbf{Pol}^K \ : \ P \vDash F \right\}$ for some set of formulas F.

Then if $P, Q \in \mathcal{C}$, then $P \sqcup Q \vDash F$.

If $P \in \mathcal{C}$ and $P \looparrowright X$, then suppose $X \notin \mathcal{C}$, i.e. there exists a formula $\varphi \in F$ such that $X \nvDash \varphi$. In other words, there exists an evaluation $\nu : \text{Fmla} \to X$ and a point $x \in X$ such that $X, x \nvDash \varphi$. We may select a triangulation S of X such that $\nu(\varphi)$ is a subtriangulation of S. Now call Σ the triangulation of X realizing the face up-reduction $P \looparrowright X$.

From [**?**] we know that there exists a barycentric subdivision of Σ triangulating S.

Furthermore, we can extend any up-p-morphism $P \to \Sigma$ to an up-p-morphism $P \to \Sigma^{(k)}$ for any $k \in \mathbb{N}$.

This means that we have an up-p-morphism $P \to \Sigma^{(k)} \to S$, therefore $P \nvDash \varphi$, leading to a contradiction. □

Proof ($\Longleftarrow$). Let $\mathcal{C}$ be a class of polyhedra closed under (I) and (II), call $\mathcal{L} := \text{Log}(\mathcal{C})$ and call $\mathcal{C}' := \left\{ P \in \mathbf{Pol}^K \ : \ P \vDash \mathcal{L} \right\}$

Clearly $\mathcal{C} \subseteq \mathcal{C}'$. Let $X \in \mathcal{C}'$, we want to show that $X \in \mathcal{C}$.

We know that $X \vDash \mathcal{L}$, and let $\Sigma \in \text{Tr}(P')$.

Call v_i the vertices of Σ. We know that $\bigsqcup_i (\uparrow v_i) \twoheadrightarrow \Sigma$ is a p-morphism.

Consider $\chi_i := \chi(\uparrow v_i)$ the Jankov-Fine formula for the (rooted) subposet.

As $(\uparrow v_i) \nvDash \chi_i$, we know that $\Sigma \nvDash \chi_i$, as all $(\uparrow v_i)$ are generated subframes of Σ.

In particular $X \twoheadrightarrow \Sigma$, thus $X \nvDash \chi_i$, meaning none of the χ_i are elements of $\mathcal{L}$.

This means that for each χ_i there exists some $P_i \in \mathcal{C}$ such that $P_i \nvDash \chi_i$, thus $P_i \twoheadrightarrow (\uparrow v_i)$.

This means that $P := \bigsqcup_i P_i$ admits an up-p-morphism to $\bigsqcup_i (\uparrow v_i)$, and we can take the composition with $\bigsqcup_i (\uparrow v_i) \twoheadrightarrow \Sigma$ to find $P \looparrowright X$. By (I) we know $P \in \mathcal{C}$, and by (II) $X \in \mathcal{C}$. □

2 Goldblatt-Thomason II

Now we want to show that (II) is equivalent to

(II') $P \in \mathcal{C}$ and there exists an open polyhedral map $P \supseteq U \to X$ for U an open subpolyhedron of P, then $X \in \mathcal{C}$.

It's clear that (II') implies (II) as X admits an open polyhedral map $X \twoheadrightarrow \Sigma$ to any of its triangulations. It is less clear why (II) implies (II')

We will split this proof into two separate parts.

Lemma 3. *Let $f : K \to K'$ be a map of simplicial complexes. Then f is open if and only $|f|$ is open.*

Proof. Recall that if $|\Sigma|$ is the realization of Σ then there exists an open polyhedral map $\pi : |\Sigma| \to \Sigma$, allowing us to draw the diagram

$$\begin{tikzcd} {|\Sigma'|} \arrow[r, dashed, two heads] \arrow[d, two heads, "\pi'"'] & {|\Sigma|} \arrow[d, two heads, "\pi"] \\ \Sigma' \arrow[r] & \Sigma \end{tikzcd}$$

One side of the implication is simple: If $|f|$ is open, then suppose $U' \subseteq \Sigma'$ is an open subcomplex, i.e. U' is upward-closed. $V' = \pi'^{\leftarrow}(U)$ is an open subpolyhedron of $|\Sigma'|$.

$f(U) = \pi|f|(V)$ is open.

Conversely, suppose $V' \in |\Sigma'|$ is open. Then by [**?**] there exists an iterated baricentric subdivision $\Sigma'^{(k)}$ of Σ' extending Σ' and covering V'.

We may extend f to $\Sigma'^{(k)}$, and by the previous lemma it will be open. We may do the same to $|f|(V)$, thus $|f|(V) = \pi^{(l)\leftarrow} f^{(k)}(\pi^{(k)}(V)$ which is open. □

2.1 Nerving

This is the correct moment to fix some notations about nerves as the following lemmas will use it extensively.

Definition 4. *Let P be a compact polyhedron, $\Sigma \in \mathrm{Tr}(P)$. Then $N(\Sigma)$ is the poset consisting of chains $(\sigma_0 < ... < \sigma_k)$ for $\sigma_i \in \Sigma$ ordered by inclusion. This constitutes the face poset of a simplicial complex triangulating P. We call $\overline{\sigma}$ elements of $N(\Sigma)$ and in particular if $\sigma_i \in \Sigma$, then $\overline{\sigma}_i = (\sigma_i)$ (the chain with one element containing it).*

When taking iterated nerves we say $\overline{\sigma}^1 = \overline{\sigma}$ and $\overline{\sigma}^{n+1} = \overline{\overline{\sigma}^n}$

Lemma 5. *If $S \subseteq \Sigma$ is a closed subcomplex of Σ, then $N(S) \subseteq N(\Sigma)$ is a subcomplex of $N(\Sigma)$. Furthermore S and $N(S)$ triangulate the same closed subpolyhedron of P*

Proof. $S \subseteq \Sigma$ is a subcomplex if and only if it's downward closed. Let $\sigma \in \Sigma$, then $\overline{\sigma} = (\sigma) \in N(S)$ If $\overline{\tau} \in N(S)$, then $\overline{\tau} = (\tau_0 < ... < \tau_k)$ for $\tau_0, ..., \tau_k \in S$. This means that any $(\tau_i < ... < \tau_j) \in \downarrow \overline{\tau}$ is also comprised of elements of S, thus a member of $N(S)$.

Moreover recall that $|N(S)| = |S|$ if S is a simplicial complex. We call $N(S) := \overline{S}$ following the notation described above. □

Proposition 6. *Let X be a compact polyhedron, $\Sigma \in \mathrm{Tr}(X)$. Call $\pi : X \to \Sigma, \pi_1 : X \to N(\Sigma)$ the maps onto the triangulation and its nerve respectively.*

Then for all $x \in X$, $\overline{\pi(x)} \leq \pi_1(x)$ in $N(\Sigma)$.

Proof. As $N(\Sigma)$ is a baricentric subdivision $\overline{\pi(x)}$ is the baricenter of $\pi(x)$, which is always a vertex of $\pi_1(x)$. □

Lemma 7. *If U is an open subcomplex of Σ,*

We know that there exists a polyhedral open map $\pi : |\Sigma| \twoheadrightarrow \Sigma$, then consider $\pi^{\leftarrow}(U) := |U| \subseteq |\Sigma|$ Call $\pi_1 : |\Sigma| \twoheadrightarrow N(\Sigma)$ the open polyhedral map onto the nerve. Then $\sigma \in \pi_1(|U|)$ if and only if $\sigma \in \uparrow \overline{\sigma}_i$ for some $\sigma_i \in U$

Proof. Note that if $\sigma \in \pi_1(|U|)$, this means that there exists a point x in $|U|$ such that $\pi_1(x) = \sigma$.

As $|U| = \pi^{\leftarrow}(U)$, this means that $\pi(x) \in U$.

Then $\overline{\pi(x)}$ is a simplex of $N(\Sigma)$, and $\overline{\pi(x)} \leq \pi_1(x)$, i.e. $\pi_1(x) \in \uparrow \overline{\pi(x)}$.

Conversely, if $\sigma \in \uparrow \overline{\sigma_i}$ for some $\sigma_i \in U$, then $\sigma = (\sigma_i < s_1 < ... < s_k)$. As U is upward-closed, each s_i is an element of U as well. This means each $\overline{s_i}$ is a vertex of σ. $\pi_1^{\leftarrow}(\overline{s_i})$ is the center of a simplex in U, thus in particular $\pi_1^{\leftarrow}(\sigma) \in |U|$.

CHECK!!

□

Definition 8. *The previous lemma justifies the notation*

$$N(U) := \bigcup_{\sigma \in U} \uparrow \overline{\sigma}$$

for any open subcomplex $U \subseteq \Sigma$ of a simplicial complex Σ.

Proposition 9. *Let X be a compact polyhedron, U an open subpolyhedron and $C \subseteq U$ a closed subpolyhedron. Let $\Sigma \in \mathrm{Tr}(X)$ be a triangulation triangulating U and C. as Σ_U and Σ_C*

Let $B = \downarrow\uparrow \overline{\overline{\Sigma_C}}^2 \subseteq N^2\Sigma$

Then $|B| \subseteq U$.

Proof. We showed above that $|B| \subseteq U \iff B \subseteq N^2(\Sigma_U)$.

1) $\Sigma_C \subseteq \Sigma_U$, thus $\overline{\Sigma}_C \subseteq N(\Sigma_U)$, thus $\overline{\Sigma}_C^2 \subseteq N^2(\Sigma_U)$

2) $\uparrow \overline{\Sigma}_C^2 \subseteq N^2(\Sigma_U)$ as $N^2(\Sigma_U)$ is upward-closed.

3) We want to show that $\downarrow\uparrow \overline{\Sigma}_C^2 \subseteq N^2(\Sigma_U)$. Let $x \in \downarrow\uparrow \overline{\Sigma}_C^2$, we want to show that $x \in \uparrow \overline{\sigma}^2$ for some $\overline{\sigma} \in N(\Sigma_U)$

Write $x = (\overline{\sigma}_0 < ... < \overline{\sigma}_k)$.

We know that $x \subseteq (\overline{c} < \overline{\sigma}_0 < ... < \overline{\sigma}_k)$ for some $\overline{c} \in \overline{\Sigma}_C$, as it's in the closure of $\uparrow \overline{\Sigma}_C^2$.

This means that $\overline{\sigma}_0 \in \uparrow \overline{c}$, and $c \in \Sigma_C \subseteq \Sigma_U$.

This implies that $\overline{\sigma}_0 \in N(\Sigma_U)$, which in turn implies $x \in \uparrow \overline{\sigma}_0^2$ is an element of $N^2(\Sigma_U)$. □

Proposition 10. *Let P, X be compact polyhedra, T a triangulation of P, Σ a triangulation of X and $f : T \twoheadrightarrow \Sigma$ a p-morphism. Then there exists a surjective up-p-morphism $F : P \twoheadrightarrow X$.*

Proof. Let v_i be vertices of Σ. As v_i are closed simplices, $f^{\leftarrow}(v_i)$ must contain vertices. Choose a vertex p_i in each $f^{\leftarrow}(v_i)$.

Call $U := \bigcup_i \uparrow_{N^2}\downarrow\uparrow_N (\overline{p}_i)$ and define $F : |U| \to X$ as the geometric realization of $\tilde{f}_{|U}$.

We want to show that $\tilde{f}$ is open and surjective:

It is open since it's the restriction of an open map to an open subset.

Note that

$$\tilde{f}(U) = \bigcup_i \uparrow_{N^2}\downarrow\uparrow_N N(f)(\overline{p_i}) = \bigcup_i \uparrow_{N^2}\downarrow\uparrow_N \overline{v}_i$$

We can show that $\bigcup_i \downarrow\uparrow_N \overline{v}_i$ covers $N(\Sigma)$: Let $\overline{\sigma}$ be a simplex of $N(\sigma)$.

Then $\overline{\sigma} = (\sigma_0 < ... < \sigma_k)$. If $\sigma_0 = v_i$ is a vertex, then $\overline{\sigma} \in\uparrow v_i \subseteq\downarrow\uparrow v_i$. If σ_0 is not a vertex, then consider any $w \in V_{\sigma_0}$: $w < \sigma_0$, thus

$(\sigma_0 < ... < \sigma_k) \leq (w < \sigma_0 < ... < \sigma_k)$ thus $\overline{\sigma} \in\downarrow (\uparrow w)$.

As $\downarrow\uparrow v_i$ cover Σ, $\uparrow\downarrow\uparrow v_i \supseteq\downarrow\uparrow v_i$ cover Σ as well, and form an open cover, thus U is open, thus $\tilde{f}$ is open.

by the previous lemma F is open since $\tilde{f}$ is. □

Proposition 11. *Let P, X be compact polyhedra, Σ a triangulation of X and $f : P \twoheadrightarrow \Sigma$ an up-p-morphism defined on an open subpolyhedron $U \subseteq P$.*

Then there exists a triangulation T of P and a closed subcomplex T_U such that

- $|T_U| \subseteq U$
- $f|_{T_U} : T_U \to \Sigma$ *is a surjective p-morphism*

Proof. Following the steps of the previous proof, choosing $T_U :=\downarrow\uparrow_{N^2}\downarrow\uparrow_N p_i$ in an appropriate triangulation is enough. To ensure that $|T_U| \subseteq U$ we need to choose a triangulation T such that p_i are vertices and it triangulates f. Then pick $\tilde{T} = N^2T$

By proposition 9, p_i is a closed subcomplex in T_U, thus $\uparrow\downarrow p_i$ is a closed subcomplex contained in $N^2(T_U)$, thus $\uparrow\downarrow (\uparrow\downarrow p_i)$ is a closed subcomplex contained in $N^2(N^2(T_U)) = N^4(T_U) \subseteq N^2\tilde{T}$,

then we apply the previous proposition to it. □